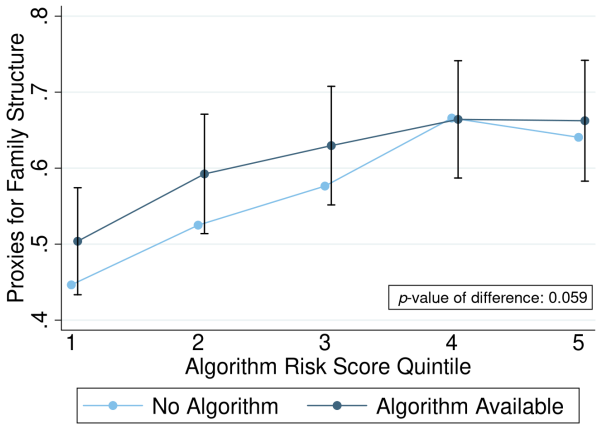
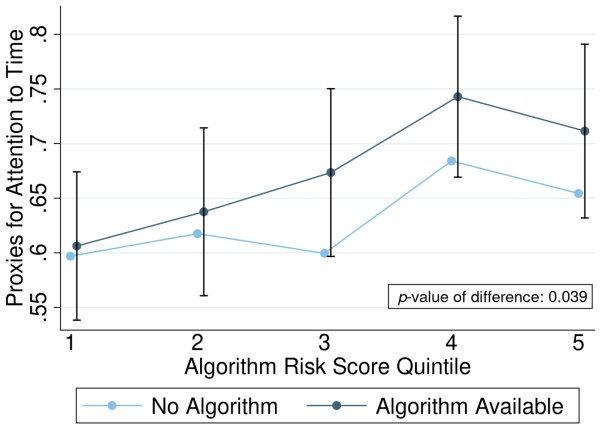
historical information. Such attention to time is about understanding current events in light of families' history. Overall, discussion of time-related proxies increased by 4.3 percentage points or 6.5 percent ($p = 0.039$, Appendix Table A.13). Discussions seemed to have refocused toward referral elements that algorithm support made more salient.

Figure 4: Access to the Algorithm Changed Worker Discussion Content

(i) Attention to Family Structure



(ii) Attention to Time



Notes: This figure illustrates changes in worker discussion content toward topics related to the design of the algorithm. Children are grouped by algorithm score quintiles on the horizontal axis. The vertical axis reports probabilities of whether workers mentioned a proxy for family structure (Panel I) or a proxy for time (Panel II). Proxies for family structure include mentioning multiple children (terms: sibling, other child, children, kids, sib, brother, sister, older child, younger child) and discussing custody (terms: custody, co parenting, co-parenting). Proxies for attention to time include referring to the urgency of the call (terms: immediate, emergency, imminent, urgen), mentioning the past (terms: last, hx [history], [name of state record-keeping software], past, prior, previous, histor, cw [past child welfare], dhs [past department of human services], same concern), and referring to a new situation (terms: new, current, now, sober). 90% confidence intervals are reported and randomization controls are included throughout.

5.4 Human-plus-Algorithm Gains under Complexity

Ehrmann et al. (2022) view cognitive load as a critical determinant of whether algorithmic tools are complementary to human expertise. Complicated interfaces or unnecessary alerts from algorithmic decision supports could contribute to cognitive overload. In contrast, a machine learning tool designed in collaboration with users could reduce what cognitive theory calls extraneous (modifiable) cognitive load (Sweller, 2011). In our setting, having an algorithm aggregate vast, disorganized historical data could allow workers to reallocate effort to other inherently difficult parts of their decision task (intrinsic cognitive load).

According to cognitive load theory, one would expect greater improvements in welfare when cases are complex and where workers are more likely to benefit from the algorithm making family history more digestible. In Table 3, we consider heterogeneity in algorithm benefits under three distinct indicators of

complexity: (1) discussion notes that mention uncertainty about a child's situation or how to address it, containing expressions such as "?", "unknown", "unclear", and "uncertain" (column 1), (2) decisions that take longer than 15 minutes, corresponding to the top quartile of decision length (column 2), and (3) children with a recent prior potential-maltreatment-related hospitalization (column 3). Each of the three measures captures a distinct dimension of complexity: the largest correlation coefficient between any two complexity measures is $\rho = 0.16$. Algorithm support did not affect the frequency of uncertainty-related text markers nor the probability of a long discussion, either in levels or when interacted by score, and the third proxy for complexity was exogenous to tool availability.

Table 3: Algorithm Benefits were Largest for Complex Referrals

Child Harm after Randomization, by Type of Complexity

	Outcome: Harm Index Type of Complexity: Column Titles		
	(1) Text Marker Uncertainty	(2) Long Discussion $\geq$ 15 min	(3) Any Index Hospital Visit, Previous 180 Days
Algorithm Available	0.001 (0.025)	-0.008 (0.024)	-0.001 (0.016)
Type	0.088** (0.045)	0.127*** (0.045)	0.552*** (0.119)
Algorithm Available · Type	-0.097** (0.049)	-0.098* (0.053)	-0.314** (0.125)
Randomization Controls	Yes	Yes	Yes
Frequency of Type	0.519	0.266	0.157
Observations	4,681	3,754	4,681

Notes: This table presents impacts of the algorithmic tool on child harm across three measures of decision complexity. The outcome in each regression is the standardized harm index of potential maltreatment-related hospitalizations occurring at least 30 days after randomization. Column 1 reports effects of the algorithm by any markers of worker uncertainty in the decision discussion notes (any instance of "?", "unknown", "unclear", "uncertain", or a related word). Column 2 reports effects of the algorithm by whether workers discussed the referral for at least 15 minutes (top quartile of discussion time), estimated on the subset of referrals with well-measured timestamps. Column 3 reports effects of the algorithm by whether a child had a recent potential-maltreatment-related hospital visit (high priority visit, injury, avoidable ER, maltreatment ICD) in the 180 days prior to the referral. The means at the bottom of the table show the frequency of each of these three binary measures of complexity. Randomization controls are included throughout. Standard errors are clustered at the household level. Significance reported as: * p<0.1, ** p<0.05, *** p<0.01.

Consistent with theory, gains from algorithm support were greatest among complex referrals. Across all three complexity measures, reductions in harm from the algorithmic tool were almost exclusively concentrated among complex referrals (Table 3, see rows "Algorithm Available · Type" and "Algorithm Available"). The findings also hold when controlling for algorithm risk score. Although each of the three complexity measures are correlated with above-average levels of harm in the control group (Table 3, row "Type"), these findings are consistent with recent work showing that algorithm support is particularly beneficial to

individuals for whom a task is complex (Brynjolfsson et al., 2023).

Did the algorithmic tool's informational content (risk score) matter, or did its presence merely serve as a behavioral reminder or nudge to workers? Although difficult to answer definitively, the fact that workers regularly checked and recorded the algorithm score – or lack thereof – in both the intervention and control groups (Appendix Figure A.4), together with score-based heterogeneity in discussion content and decisions (Figures 2 and 4) suggest that benefits of the algorithm are at least partially attributable to the process of reviewing the predicted risk scores. Although the specific 1-to-20 value of the algorithm's risk score likely helped workers allocate their attention more effectively, we cannot definitively rule out benefits from a general increase in the salience of family history – or other characteristics – upon seeing a listed score, regardless of its magnitude. An important topic for future research is understanding the impacts of how information is presented to users in similar settings.

6 Human-plus-Algorithm vs. Algorithm-Only Counterfactuals

To what extent do human decision-makers contribute their independent expertise to improve upon algorithms? As machine learning decision supports become more widespread, so do questions about human experts' contributions to decisions, and when to automate. In this study, outcomes improved when workers were given access to an algorithm. Workers, however, did not target investigations toward children with higher algorithm risk scores, but instead used complementary information to target more effectively on other dimensions. Given how well the algorithm risk score predicts relevant outcomes (Appendix Figure A.3), it is reasonable to ask: Could outcomes have improved *even more* if workers had based decisions more closely on the algorithm risk score? And would there have been implications for equity?

In this section, we conduct two innovative counterfactual exercises to estimate the impact of deferring to an algorithm to make decisions. We assess whether child harm and equity would have improved under (i) investigation decisions determined solely by algorithm predictions, and (ii) marginal increased reliance on algorithm predictions. This methodology can be applied in other experimental settings to assess whether users underutilize algorithmic information. Parts of the methodology can also be applied to study hypothetical, alternative decision rules beyond the implemented algorithm.